

University Interscholastic League
2024 – 2025 Elementary Number Sense Test A

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.
Stop – Wait for Signal!

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| <p>(1) $22 + 41 =$ _____</p> <p>(2) $37 + 48 =$ _____</p> <p>(3) $108 - 42 =$ _____</p> <p>(4) $312 \div 3 =$ _____</p> <p>(5) $26 \times 5 =$ _____</p> <p>(6) $425 - 332 =$ _____</p> <p>(7) $19 + 7 + 13 =$ _____</p> <p>(8) $52 \times 11 =$ _____</p> <p>(9) $16 \times 50 =$ _____</p> <p>*(10) $2025 + 224 =$ _____</p> <p>(11) $16 \times 25 =$ _____</p> <p>(12) Which digit is in the hundredths place in 34917.20568? _____</p> <p>(13) $22 \times 18 =$ _____</p> <p>(14) 83670.2874 rounded to the tenths place is _____</p> <p>(15) What is the remainder for $60318 \div 9$? _____</p> <p>(16) The number of even whole numbers between 13 and 25 is _____</p> <p>(17) $3 \times 10^{-2} + 2 \times 10^0 + 4 \times 10^{-1} =$ _____ (decimal)</p> <p>(18) $17 \times 3 + 17 \times 7 =$ _____</p> <p>(19) DLX = _____ (Arabic Numeral)</p> | <p>*(20) $225 \times 399 =$ _____</p> <p>(21) $16 - 8 \div 2 =$ _____</p> <p>(22) $9 + 12 + 15 + 18 =$ _____</p> <p>(23) $\frac{1}{12}$ hour = _____ minutes</p> <p>(24) $123 \times 20 =$ _____</p> <p>(25) $\frac{5}{14} - \frac{3}{14} =$ _____</p> <p>(26) $99 \times 98 =$ _____</p> <p>(27) $38\% =$ _____ common fraction</p> <p>(28) If 15 ♥ costs 25¢, then 45 ♥ cost _____ ¢</p> <p>(29) $21 \div \frac{3}{10} =$ _____</p> <p>*(30) $499 \times 719 + 59 =$ _____</p> <p>(31) \$28 = _____ quarters</p> <p>(32) The largest prime number between 40 and 60 is _____</p> <p>(33) 64 ounces = _____ quarts</p> <p>(34) $3434 \div 34 =$ _____</p> <p>(35) $83\frac{1}{3}\% =$ _____ common fraction</p> <p>(36) The LCM of 24 and 16 is _____</p> <p>(37) $33 \times 12 =$ _____</p> |
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- (38) If 12 apples cost \$2.40, then 8 apples cost \$_____
- (39) 6 is to 20 as 3 is to _____
- *(40) $213 \times 667 =$ _____
- (41) $\frac{1}{8} + \frac{3}{8} =$ _____ (common fraction)
- (42) $16^2 - 6 =$ _____
- (43) If $f(x) = 3x - 5$, then $f(4) =$ _____
- (44) The radius of a circle with a circumference of 100π is _____
- (45) $2^4 \times 3^0 =$ _____
- (46) $2\frac{1}{4} \times 6\frac{1}{4} =$ _____ (mixed number)
- (47) What is the area of a square with perimeter 36? _____
- (48) $44\frac{4}{9}\% \times 18 =$ _____
- (49) If there are 12 grapes to an ounce, then 1 pound equals how many grapes? _____
- *(50) $249\frac{1}{90} \times 164 =$ _____
- (51) What is the number, **k**, in the sequence:
1, 3, 5, **k**, , 9, 11, . . . ? _____
- (52) $9\frac{4}{5} - 6\frac{9}{10} =$ _____ (mixed number)
- (53) $\sqrt{576} =$ _____
- (54) A square with area 9 is located inside a square with perimeter 36. What is the area between the squares? _____
- (55) 32 (base 3) = _____ (base 10)
- (56) The number of elements in $\{2, 4, 6\} \cup \{1, 2, 3, 4\}$ is _____
- (57) What is the perimeter of a regular hexagon with side $4\frac{1}{2}$? _____
- (58) $75 \times 32 =$ _____
- (59) $2\frac{1}{2} + 2\frac{1}{2} \times 4\frac{1}{2} =$ _____
- *(60) 160 inches = _____ centimeters
- (61) 1001 (base 2) = _____ (base 4)
- (62) $125 \times (-4) =$ _____
- (63) What is the probability a getting a sum of 9 when rolling a pair of dice? _____
- (64) 3 pints = _____ ounces
- (65) $57^2 =$ _____
- (66) $10^5 \div 6$ has remainder of _____
- (67) The area of a rhombus with diagonals $2\frac{1}{2}$ and 20 is _____
- (68) If $-3x + 15 < 54$, then $x >$ _____
- (69) $(-2.25) \times (-40) =$ _____
- *(70) $25^3 =$ _____
- (71) The multiplicative inverse of $6\frac{3}{7}$ is _____
- (72) $6^2 - 26^2 =$ _____
- (73) 125% of 440 = _____
- (74) What is the area of a trapezoid with bases $4\frac{1}{4}$, $6\frac{1}{4}$ and altitude to the bases of 20? _____
- (75) $24^2 + 6^2 =$ _____
- (76) $(1 + 3 + 5 + \dots + 11)^2 =$ _____
- (77) What is the total surface area of a rectangular box with edges 8, 6 and 5? _____
- (78) The area of a right triangle with legs $4\frac{1}{4}$ and 16 is _____
- (79) $375 \times 80 =$ _____
- *(80) 15 square miles = _____ acres

2024 – 2025 University Interscholastic League Elementary Number Sense Test A – Key

(1) 63	*(20) 85287 – 94263	(38) 1.60	(59) $13\frac{3}{4}$; 13.75; $\frac{55}{4}$
(2) 85	(21) 12	(39) 10	*(60) 387 – 426
(3) 66	(22) 54	*(40) 134968 – 149174	(61) 21
(4) 104	(23) 5	(41) $\frac{1}{2}$	(62) -500
(5) 130	(24) 2460	(42) 250	(63) $\frac{1}{9}$
(6) 93	(25) $\frac{1}{7}$	(43) 7	(64) 48
(7) 39	(26) 9702	(44) 50	(65) 3249
(8) 572	(27) $\frac{19}{50}$	(45) 16	(66) 4
(9) 800	(28) 75	(46) $14\frac{1}{16}$	(67) 25
*(10) 2137 – 2361	(29) 70	(47) 81	(68) -13
(11) 400	*(30) 340898 – 376782	(48) 8	(69) 90
(12) 0	(31) 112	(49) 192	*(70) 14844 – 16406
(13) 396	(32) 59	*(50) 38796 – 42879	(71) $\frac{7}{45}$
(14) $83670.3; 83670\frac{3}{10};$ $\frac{836703}{10}$	(33) 2	(51) 7	(72) -640
(15) 0	(34) 101	(52) $2\frac{9}{10}$	(73) 550
(16) 6	(35) $\frac{5}{6}$	(53) 24	(74) 105
(17) 2.43	(36) 48	(54) 72	(75) 612
(18) 170	(37) 396	(55) 5	(76) 1296
(19) 560		(56) 5	(77) 236
		(57) 27	(78) 34
		(58) 2400	(79) 30000
			*(80) 9120 – 10080

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.